December 1, 2006

Physical Cumes (2006)

Equilibrium Electrochemistry

In equilibrium electrochemistry, quantitative calculations of electrochemical quantities are all based on equilibrium thermodynamics.

(1) **(20 points)** The starting point of quantitative equilibrium electrochemistry from equilibrium thermodynamics is Equation 1.

$$W_{e,\,max} = \Delta G \qquad (1)$$

Please define both parameters. Is Equation 1 a special form of the second law of thermodynamics, or a variation of the first law of thermodynamics, or not related to either of the two laws mentioned?

(2) **(30 points)** From Equation 1, please deduce Equation 2. Here, ν is the number of electrons transferred in a stoichiometric redox reaction, F is Faraday constant, E is the corresponding cell potential of the redox reaction, and $\Delta_r G$ is reaction free energy of the redox reaction.

$$-\nu FE = \Delta_r G \qquad (2)$$

(3) **(30 points)** From Equation 2, you should be able to deduce the famous Nernst equation. Please show it. It is allowed to directly apply thermodynamic equations regarding chemical equilibrium.

$$E = E^\circ - RT/\nu F \ln Q \qquad (3)$$

(4) **(20 points)** Faraday constant is a very important constant in equilibrium electrochemistry. Please calculate it from the values of the physical constants listed below.

Speed of light: 3.00×10^{8} (m s^{-1})
Elementary charge: 1.60×10^{-19} (C)
Boltzmann constant: 1.38×10^{-23} (J K^{-1})
Gas Constant: 8.31 (J K^{-1} mol^{-1})
Mass of electrons: 9.12×10^{-31} (kg)